

Fertilizer & Nutrient Management

AgriMitra AI Knowledge Base — Document 2 of 6 — Nutrient Guidance by Crop

General Principles

- Always base fertilizer decisions on a recent soil test where possible — this document gives general, commonly recommended ranges only
- Nitrogen (N) drives vegetative growth and leaf color; Phosphorus (P) supports root and flower development; Potassium (K) improves fruit/grain quality and disease resistance
- Apply Nitrogen in split doses (basal + 1–2 top dressings) to reduce leaching loss and improve uptake efficiency
- Combine chemical fertilizers with organic sources (farmyard manure, compost, green manure) to maintain long-term soil health
- Micronutrient deficiencies (Zinc, Boron, Iron) are increasingly common in intensively cropped soils — a soil test will indicate if correction is needed

Organic vs Chemical Fertilizers

Aspect	Organic (FYM, compost, vermicompost)	Chemical (Urea, DAP, MOP)
Nutrient release	Slow, sustained over the season	Fast, immediately available
Soil health impact	Improves structure & microbial activity	No structural benefit; overuse can harm soil biology
Cost per season	Lower recurring cost, more labor	Higher recurring cost, less labor
Best use	Basal application, long-term soil building	Top dressing for quick-response nutrient needs

Indicative Stage-wise Nutrient Schedule (per acre)

Figures below are commonly cited general ranges for reference only — actual recommendations vary by soil test, variety, and state guidelines.

Paddy (Rice)

Stage	N	P2O5	K2O
Basal (at transplanting)	1/3 of total N	Full dose	1/2 of total K
Active tillering (20–25 DAT)	1/3 of total N	—	—
Panicle initiation	1/3 of total N	—	1/2 of total K

Cotton

Stage	N	P2O5	K2O
Basal (at sowing)	1/4 of total N	Full dose	1/2 of total K
30–40 days (squaring)	1/4 of total N	—	—

Stage	N	P2O5	K2O
60–70 days (flowering)	1/4 of total N	—	1/2 of total K
90 days (boll development)	1/4 of total N	—	—

Chilli

Stage	N	P2O5	K2O
Basal (at transplanting)	1/4 of total N	Full dose	1/2 of total K
30 days after transplanting	1/4 of total N	—	—
Flowering	1/4 of total N	—	1/2 of total K
Fruit development	1/4 of total N	—	—

Maize

Stage	N	P2O5	K2O
Basal (at sowing)	1/3 of total N	Full dose	Full dose
Knee-high stage (25–30 days)	1/3 of total N	—	—
Tasseling stage (45–50 days)	1/3 of total N	—	—

Groundnut

Stage	N	P2O5	K2O / Gypsum
Basal (at sowing)	Full dose (modest, legume)	Full dose	Full K dose
Flowering / pegging (35–40 days)	—	—	Gypsum top-dress for pod development

Sugarcane

Stage	N	P2O5	K2O
Basal (at planting)	1/4 of total N	Full dose	1/2 of total K
Tillering (45–60 days)	1/4 of total N	—	—
Grand growth (90–120 days)	1/4 of total N	—	1/2 of total K
Post grand-growth top dress	1/4 of total N	—	—

Common Nutrient Deficiency Symptoms

Nutrient	Visible Symptom	Commonly Affected Crops
Nitrogen (N)	Uniform yellowing of older/lower leaves; stunted growth	All crops, especially cereals
Phosphorus (P)	Dark green/purplish leaves, poor root and flower development	Maize, groundnut, young seedlings
Potassium (K)	Yellowing/scorching of leaf margins, weak stalks, poor grain fill	Cotton, sugarcane, paddy
Zinc (Zn)	Interveinal yellowing (khaira disease in paddy), stunted growth	Paddy, maize
Boron (B)	Poor flower/fruit set, hollow stem in some crops	Cotton, groundnut, chilli
Iron (Fe)	Yellowing between veins on young/upper leaves	Groundnut in alkaline soils

Note for verification: N-P-K figures here are indicative ranges commonly used for reference, not exact prescriptions. Always confirm exact per-acre dosage against a current soil test report or your state agriculture department / ICAR recommendation before advising a farmer.